

Homework 5: Countability & Decidability

Due:04/08/2026

This assignment is due on **11:59 PM, April 8, 2026**. You may turn it in up to 48 hours late, but assignments turned in by the deadline receive 3% extra credit.

You should submit a typeset or *neatly* written pdf on Gradescope. The grading TA should not have to struggle to read what you've written; if your handwriting is hard to decipher, you will be asked to typeset your future assignments.

You may collaborate with other students, but any written work should be your own.

1. Countability (4 points - Both Sections)

Let P be the set of all polynomials with positive integer coefficients. An element of P is any function $f(x) = a_n x^n + a_{n-1} x^{n-1} \dots + a_1 x + a_0$ where $n \geq 0$ and all coefficients $a_i \in \mathbb{Z}^+$.

Prove that P is countably infinite.

Answer: Consider the set $A = A_0 \cup A_1 \cup \dots$ where A_i is the set defined by the polynomials with degree i (highest power is i). So note that for any element of P i.e a polynomial as described above p can be put uniquely into one of these sets based on the degree of p , so the sets $A_0 \dots$ partition the set P . Now note that each A_i is countable. We see this because the possibly polynomials we can define are by enumerating each $a_0, \dots, a_i$, as each can take $\mathbb{Z}^+$ (note a_i takes $\mathbb{Z}^+ \setminus \{0\}$ to be more specific) so the possibly polynomials with degree i are $\mathbb{Z}^+ \times \dots \times \mathbb{Z}^+ \times \mathbb{Z}^+ \setminus \{0\}$ or $(\mathbb{Z}^+)^i \times \mathbb{Z}^+ \setminus \{0\}$. Now note that this is a countable set be the finite Cartesian product of countable sets are countable (specifically we can use a diagonal argument to map them into the naturals). Now note that as each A_i is countably infinite and we have $A = A_0 \cup A_1 \dots$ i.e A is the countable union of countably infinite sets, we have that A is countable as well (again we can use a diagonal argument to show that a countably infinite union of countably infinite sets are countably infinite).

2. Uncountability (4 points - Section B)

Let P be a countably infinite set of planets in the universe (e.g., $S = p_1, p_2, p_3, \dots$).

Let C_P be the set of all possible ways to define connections (relationships, wormholes, hyperspace routes, etc.) between these planets. A "connection map" $C \in C_P$ is simply a specification of which pairs of planets are connected.

Use diagonalization to show that there are no surjections from the countable set of planets P to the set of all possible connection maps C_P .

(Hint: consider a table like the ones we made in class, where the left side is all the vertices $v_1, v_2, \dots$ in V and the top row is all the graphs $G_1, G_2, \dots$ in G_V . What condition should you check for in the cell corresponding to (v_i, G_j) that will allow you to diagonalize the table and form a new graph?)

Answer:

3. **Decidability** (4 points - Section B)

Let L_1 and L_2 be two decidable languages, and let L_3 be a language which is Turing-recognizable but not decidable. “ $-$ ” denotes set subtraction.

(2 points) Must $L_3 - (L_1 \cap L_2)$ be Turing-recognizable? Prove your answer.

Answer:

(2 points) Must $L_1 - (L_2 \cup L_3)$ be Turing-recognizable? Prove your answer. (Hint: Σ^* is decidable.)

Answer:

4. **Undecidability** (4 points - Both Sections)

Show that the following languages are undecidable. Do not use Rice’s theorem as we do not cover it in the lectures.

(2 points) $L_3 = \{\langle M \rangle \mid 00110 \in L(M)\}$

Answer: We know that if A is reducible to B then if A is undecidable then B is as well. So it is enough to show that A_{TM} is reducible to L_3 . First assume that L_3 is decidable, decided by R . Now consider the input $A_{TM} : \langle M, w \rangle$, we will first construct a modified TM, M_1 defined as,

M_1 : on input x , does the following,

- (a) If $x = 00110$ then run M on w and return it’s output
- (b) Else we accept.

So clearly if $00110 \in L(M_1)$ implies we that M accepted w or that $\langle M, w \rangle \in A_{TM}$. Similarly if we have $\langle M, w \rangle \in A_{TM}$ then we know M accepts w and that we accept for $x = 00110$ and hence $00110 \in L(M_1)$.

This means if we can decide whether $00110 \in L(M_1)$ then we can decide whether $\langle M, w \rangle \in A_{TM}$. Now take our decider R from above, we can make the following construction for S ,

S : on input $\langle M, w \rangle$

- (a) Construct M_1
- (b) Run R on $\langle M_1 \rangle$
- (c) If R accepts ($00110 \in L(M_1)$) then accept
- (d) If R rejects then reject.

So S is a decider for A_{TM} , a contradiction. Hence, we cannot decide whether $001100 \in L(M)$ i.e L_3 is undecidable.

(2 points) $L_4 = \{(\langle M \rangle, q) \mid q \text{ is a useless state of } M\}$, where a state q of a Turing machine M is called “useless” if there is no input that causes M to enter state q .

Answer: We'll once again show that we can reduce A_{TM} to L_4 . First assume that L_4 is decidable and hence there is some decider R for it. Now take an input of A_{TM} i.e $\langle M, w \rangle$ and construct a modified TM, M_1 as follows,

M_1 : on input x does,

- (a) If $x = w$ then we simulate M on w , and on M accepting w we jump to a new state q and accept.
- (b) If $x \neq w$ then we just reject.

So essentially we have q is a useful state of M if and only if $\langle M, w \rangle \in A_{TM}$. Using this we construct S as follows,

S : on input $\langle M, w \rangle$

- (a) Construct M_1
- (b) Run R on $\langle M_1, q \rangle$ (where q is the new state we added for w)
- (c) If R accepts it implies that q is a useless state of M_1 and hence we reject
- (d) If R rejects we accept.

So here we have a decider for A_{TM} a contradiction.

5. Homomorphisms (4 points - Section X)

Let Σ and Γ be two alphabets. A homomorphism is a function $f : \Sigma^* \rightarrow \Gamma^*$ such that for all $x, y \in \Sigma^*$, $f(xy) = f(x) \circ f(y)$. (You can think of f as replacing each character in a word by a specific string in Γ^* .) Show that decidable languages are not closed under homomorphism, by giving a language L and a homomorphism f such that L is decidable but $\{f(w) \mid w \in L\}$ is not.

Answer: First consider $A_{TM} = \{\langle M, w \rangle \mid M \text{ accepts } w\}$. Now we transform this by encoding A_{TM} in binary i.e in $\{0, 1\}$. Consider the $\Sigma = \{0, 1, |, t\}$. We will construct L as follows, $L = \{x|t^n \mid \text{if } M \text{ accepts } x \text{ within } n \text{ steps}\}$ where M is a TM that recognizes A_{TM} . So note that L is decidable because for any input i.e $x|t^n$ first we check if it's of the right form and then simply simulate M for t steps on x and if it accepts we accept else we just reject. So our TM will always decide. Now define our homomorphism f such that $f(0) = 0, f(1) = 1, f(t) = \varepsilon, f(|) = \varepsilon$. Now note that we're left with $f(L) = \{x \mid \text{there is some } t \text{ such that } M \text{ accepts } x \text{ in less than } t \text{ steps}\}$. But note that this is just A_{TM} as we have the binary encoding of M, w for which M accepts w in some finite number of steps.

6. Busy Beavers (4 points - Section X)

Let $BB(k)$ be a function from $\mathbb{N} \rightarrow \mathbb{N}$ such that $BB(k) = n$ if for all Turing machines with k states that halt on input ε , n is the maximum number of 1's left on the tape when started with ε as input. Show that $BB(k)$ is not a computable function.

Answer: First let us assume that $BB(k)$ is a computable function. Hence, there must be an TM say M that computes it (assume the unary encoding so we have k is encoded as 1^k). Now let M be such that on input 1^k will leave us with a maximum of $1^{BB(k)}$

ones on the tape after halting for all k . Now define a TM , M' as follows. M' writes k 1s on the tape and then doubles it to $2k$ and then simulates M and halts with $BB(2k)$. Now to write k ones we need k states and we need some constant number of steps say m to double it and simulate M . So our total states is $m + n$ and we have $BB(m + n)$ is the max number of 1s a $m + n$ states TM halts with which implies that it halts with more 1s than the case for $2n$ state TM i.e we have $BB(n + c) \geq BB(2n)$, for every n . However, we know that BB is a strictly increasing function, we can show this by considering M an n -state TM that achieves the maximum 1's i.e M halts writing $BB(n)$ 1's on the tape. Now construct M' with $n + 1$ states such that M' simulates M first using the same n states of M and then instead of halting we jump to a new state and write a 1 where we didn't before, hence writing at least one extra one onto the tape. This means that BB is strictly increasing and hence the above inequality is a contradiction.